RetailIQ: Customer & Revenue Intelligence

Executive Summary

RetailIQ turns two years of transaction history into customer, revenue, and planning decisions. The project combines validated retail data, leakage-safe customer return modelling, weekly revenue forecasting, and a seven-page decision dashboard.

The final customer model generalized credibly. Gradient Boosting achieved 0.749 PR-AUC and 0.806 ROC-AUC on an untouched chronological test set of 14,341 customer snapshots.

Forecasting stayed deliberately simple. A 52-week seasonal naïve benchmark outperformed more complex candidates in rolling backtests and recorded 18.3% sMAPE on the frozen 16-week holdout.

Analysis-ready rows

1,055,238

Rows retained after exact duplicate removal and conservative quality handling.

Source: UCI Online Retail II cleaned analytical tables
Table: data/processed/transactions.parquet

Final customer-model PR-AUC

0.75

Precision-recall area on the untouched chronological final test set.

Source: Frozen return-risk validation and final-test evidence
Tables: reports/generated/classification_final_test_metrics.json, models/return_risk_metadata.json

Forecast holdout sMAPE

18.3%

Symmetric mean absolute percentage error on the frozen 16-week holdout.

Source: Frozen weekly revenue forecast evidence
Table: models/revenue_forecast_metadata.json

What the evidence shows

The workflow cleaned 1.07M source rows into 1.06M analysis-ready rows while preserving anonymous transactions for revenue analysis and excluding them from customer modelling. Customer features were created only from information available at each observation date, with chronological boundaries and a 90-day purge protecting against leakage.

The chart compares validation and final-test classification performance. Precision improved on the final period while recall declined, showing why campaign decisions should consider missed returners and unnecessary contacts rather than accuracy alone.

Validation and final-test classification metrics

Source: Frozen return-risk validation and final-test evidence
Tables: reports/generated/classification_final_test_metrics.json, models/return_risk_metadata.json

Metric	Score	Evaluation
Precision	0.56	Validation
Precision	0.67	Final test
Recall	0.72	Validation
Recall	0.66	Final test
F1	0.63	Validation
F1	0.66	Final test
ROC-AUC	0.78	Validation
ROC-AUC	0.81	Final test
PR-AUC	0.71	Validation
PR-AUC	0.75	Final test

Recommended next steps

1. Deploy the Streamlit dashboard as a public portfolio demonstration using the historical snapshot.
2. Validate retention actions through a randomized campaign test; predicted return likelihood is not campaign lift.
3. Add promotion, inventory, pricing, and margin data before using the revenue forecast for operational budgeting.

Caveats and assumptions

The source covers one UK-based retailer from 2009–2011, 22.8% of source rows lack customer IDs, and currency is inferred as GBP from source context. Forecast uncertainty is empirical, not guaranteed. This is a historical analytical prototype, not a live production retail system.